Saturday, July 26, 2025 11:38 AM

📡 What is SNMPwalk?

snmpwalk is a **command-line tool** used to **query and retrieve data from devices running SNMP (Simple Network Management Protocol)**. It's commonly used by network admins and penetration testers to **enumerate SNMP-enabled devices**.

🧠 In Simple Words:

snmpwalk walks through an SNMP-enabled device's internal data (like network interfaces, processes, users, uptime, and more), using a single request.

Quick SNMP basics

- **SNMP agent**: software on a device (router, switch, server) that exposes management data.
- **SNMP manager**: the client/tool that queries agents (e.g., snmpwalk, a NMS).
- **MIB** (Management Information Base): hierarchical schema that describes objects.
- **OID**: object identifier (e.g., .1.3.6.1.2.1.1.1 for sysDescr).
- **Versions**: v1, v2c (community string), v3 (user-based security: auth/privacy).
- **Ports**: UDP 161 for queries, UDP 162 for traps.

🔧 SNMPwalk Details

Feature	Info
Tool Type	SNMP client/utility
Works with	SNMPv1, SNMPv2c, and SNMPv3
Default Community	public (used for read-only access in SNMP v1/v2c)
OS Support	Linux (comes with snmp package), also works on Windows via WSL or Cygwin

📘 Basic Syntax:

```
snmpwalk -v <version> -c <community_string> <target_ip> [OID]
```

Parameter	Meaning
-v	SNMP version (1, 2c, or 3)
-c	Community string (public, private)
OID (optional)	Object Identifier to query specific info

🧪 Example Commands

🔹 **Get all available SNMP data:**

```
snmpwalk -v2c -c public 192.168.1.1
```

🔹 **Get system info only:**

```
snmpwalk -v2c -c public 192.168.1.1 1.3.6.1.2.1.1
```

This OID represents **system information** (hostname, uptime, etc.)

🔹 **SNMPwalk for interfaces:**

```
snmpwalk -v2c -c public 192.168.1.1 1.3.6.1.2.1.2.2.1.2
```

Shows all network interfaces.

🧾 Output Example:

```
SNMPv2-MIB::sysName.0 = STRING: router.local
SNMPv2-MIB::sysDescr.0 = STRING: Linux 4.19.0-17-amd64
IF-MIB::ifDescr.1 = STRING: eth0
IF-MIB::ifDescr.2 = STRING: wlan0
```

🔍 Common OIDs

OID	Description
1.3.6.1.2.1.1	System information
1.3.6.1.2.1.2.2.1.2	Network interfaces
1.3.6.1.2.1.4.20.1.1	IP addresses
1.3.6.1.2.1.25.4.2.1.2	Running processes
1.3.6.1.2.1.25.6.3.1.2	Installed software
1.3.6.1.2.1.6.13.1.3	TCP listening ports

🎯 Use Cases

Use Case	Purpose
Network Auditing	Find device info and misconfigurations
Penetration Testing	Discover hosts, services, OS, users
Device Inventory	List connected interfaces or ports
Monitoring Tools Integration	SNMPwalk used to test SNMP configurations

🛡️ **Security Considerations**

- SNMPv1/v2c is **not encrypted** – easy to sniff.
- Default community strings like public are insecure.
- Should be **disabled or restricted** on production systems.
- Always **use SNMPv3** if you need secure management.